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TRANSPORT PHENOMENA I, CHE 311

QUIZ 2

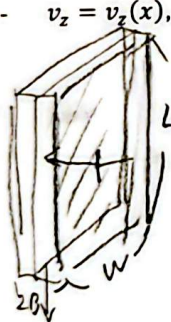
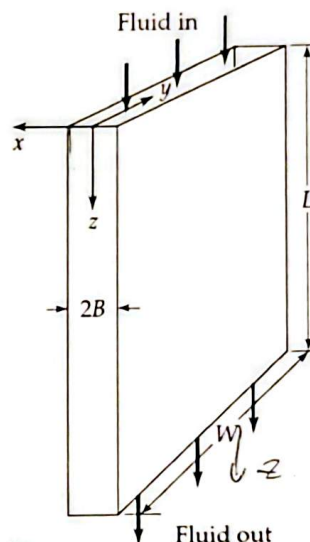
CLOSED BOOK, TIME: 15 MINUTES

A Newtonian fluid is in laminar flow through a narrow slit formed by two vertical parallel walls a distance $2B$ apart. The wall at $x = -B$ is stationary, while the wall at $x = +B$ moves with a constant velocity v_B in z direction.

Make a shell momentum balance to obtain the velocity profile $v_z(x)$.

Model assumption:

- Steady state
- Neglect edge effects, assuming $B \ll W \ll L$
- Constant density ρ and viscosity μ
- $p(z=0) = p_0$, $p(z=L) = p_L$, and $\frac{\partial p}{\partial x} = \frac{\partial p}{\partial y} = 0$
- $v_z = v_z(x)$, and $v_x = v_y = 0$.



$$\begin{aligned} \phi_{xz} &= \rho v_x v_z + p_0 + \tau_{xz} \\ &= -\mu \left(\frac{dv_z}{dx} + \frac{dv_x}{dz} \right) \end{aligned}$$



$$\begin{aligned} \phi_{zz} &= \rho v_z v_z + p_0 + \tau_{zz} \\ &= \rho v_z v_z + p - \mu \left(2 \frac{dv_z}{dz} \right) \end{aligned}$$

$$\lim_{\Delta x \rightarrow 0} \frac{LW(\phi_{xz}|_{x=x} - \phi_{xz}|_{x=x+\Delta x}) + \frac{\Delta x}{2B} \cdot W(\phi_{zz}|_{z=0} - \phi_{zz}|_{z=L}) + \rho g(\Delta x \cdot W \cdot L)}{\Delta x \cdot WL} = 0$$

$$\lim_{\Delta x \rightarrow 0} \frac{\tau_{xz}|_{x=x} - \tau_{xz}|_{x=x+\Delta x}}{\Delta x} + \frac{(p_0 - p_L) + (\rho v_z v_z|_{z=0} - \rho v_z v_z|_{z=L})}{L} + \rho g = 0$$

$$\frac{d\tau_{xz}}{dx} = \rho g$$

$$\tau_{xz} = \rho g x + C_1$$

$$\mu \frac{dv_z}{dx} = \rho g x + C_1 \quad \frac{dv_z}{dx} = \left(-\frac{\rho g}{\mu} \right) x + \left(-\frac{C_1}{\mu} \right) \quad v_z = \left(-\frac{\rho g}{\mu} \right) \frac{x^2}{2} + \left(-\frac{C_1}{\mu} \right) x + C_2$$

$$\begin{aligned} \text{at } x=B, v_z &= v_B \\ \text{at } x=-B, v_z &= 0 \end{aligned} \Rightarrow \begin{cases} v_B = \left(-\frac{\rho g}{\mu} \right) \frac{B^2}{2} + \left(-\frac{C_1}{\mu} \right) B + C_2 \\ 0 = \left(-\frac{\rho g}{\mu} \right) \frac{B^2}{2} + \left(-\frac{C_1}{\mu} \right) (-B) + C_2 \end{cases}$$

$$v_B = 2 \left(-\frac{C_1}{\mu} \right) B \quad C_1 = \left[\frac{-\mu v_B}{2B} \right] \quad \text{Q}$$

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$$C_2 = \left(-\frac{C_1}{\mu}\right)B + \left(\frac{\rho g}{\mu}\right)\frac{B^2}{2}$$

$$= \left(-\frac{1}{\mu}\right)\left(-\frac{\mu V_B}{2B}\right)B + \left(\frac{\rho g}{\mu}\right)\frac{B^2}{2}$$

$$= \frac{V_B}{2} + \left(\frac{\rho g}{\mu}\right)\left(\frac{B^2}{2}\right)$$

$$V_z(x) = \left(-\frac{\rho g}{\mu}\right)\frac{x^2}{2} + \left(\frac{V_B}{2B}\right)x + \frac{V_B}{2} + \frac{\rho g B^2}{2\mu}$$

$$\lim_{\Delta x \rightarrow 0} \left(\frac{z_{x2} - z_{x1}}{\Delta x} + \left(\frac{P_0 - P_L + \rho g L}{L} \right) \right)$$

$$\frac{dz_{x2}}{dx} = \left(\frac{P_0 - P_L}{L} \right)$$

$$z_{x2} = \left(\frac{P_0 - P_L}{L} \right)x + C_1$$

$$\tau_{xz} = -\mu \frac{dv_z}{dx} = \left(\frac{P_0 - P_L}{L} \right)x + C_1$$

$$\frac{dv_z}{dx} = \left(-\frac{P_0 - P_L}{\mu L} \right)x + \frac{C_1}{\mu}$$

$$v_z = \left(-\frac{P_0 - P_L}{\mu L} \right)\frac{x^2}{2} + \left(-\frac{C_1}{\mu} \right)x + C_2$$

$$v_z(x=B) = v_{B0}$$

$$v_z(x=-B)$$

$$v_z(x) = \left(-\frac{P_0 - P_L}{\mu L} \right)\frac{x^2}{2} + \left(-\frac{V_B}{2B} \right)x + \frac{V_B}{2} + \frac{P_0 - P_L}{2\mu L} \quad \checkmark$$